

Shiva Karthik Rallabandi

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EDUCATION

Central Michigan University

Masters of Science in Information Systems

Mount Pleasant, MI

Aug. 2024 – Dec 2025

Keshav Memorial Institute of Technology

Bachelor of Technology in Information Technology

Hyderabad, India

Aug. 2019 – May 2023

EXPERIENCE

Software Engineer 1

July 2023 – July 2024

Ivanti

Karnataka, India

- Developed and optimized distributed ITSM service management libraries using C#, .NET, and Azure Functions, refactoring microservices architecture to improve API scalability by 40% and reduce system crashes by 25%.
- Diagnosed and resolved critical system bugs using Visual Studio debugging tools and Postman API testing, enhancing system stability and reducing incident reports by 30%.
- Engineered and deployed the Trusted Agent Feature, implementing SignalR for real-time data synchronization and JWT authentication for secure access control, reducing unauthorized access attempts by 35%.
- Automated infrastructure provisioning and CI/CD workflows using Terraform and Azure DevOps, enabling rapid deployment of system testing environments and cutting deployment setup time by 50%.
- Conducted comprehensive design and code reviews, optimizing database queries in PostgreSQL and Redis, improving query response times by 45% through advanced indexing and caching strategies.

Software Engineer Intern

Jan 2023 – July 2023

Ivanti

Karnataka, India

- Created an internal cost-saving tool using C, .NET, and Azure Functions, optimizing cloud resource allocation through predictive workload analysis, reducing infrastructure costs by 25%.
- Implemented and debugged integration and system tests across Server Management Libraries, utilizing JUnit and Postman API testing, achieving 99.5% test reliability.
- Enhanced API performance and data flow in web-based applications, refactoring .NET Core microservices with asynchronous processing and optimized caching, reducing API latency by 40%.
- Strengthened application security by implementing JWT authentication and role-based access control (RBAC), reducing security vulnerabilities by 20%.
- Collaborated with cross-functional teams in an Agile environment, optimizing database transactions by 20% in MySQL and Redis, reducing query execution times and enhancing system performance.

PROJECTS

ProductiVision | *Next.js, PostgreSQL, Azure Face API, TailwindCSS, React, Node.js*

Feb 2025 – Mar 2025

- Built an AI-powered productivity tool with session-based authentication, behavioral and facial recognition for focus tracking, interactive dashboards, task-based monitoring, and AI-driven work optimization recommendations.

Fitness Buddy | *React.js, Tailwind CSS, MongoDB, Nodejs, Express, JWT*

Jan 2025 – Feb 2025

- Designed and Developed a full-stack web application for tracking workouts, connecting with fitness partners, and monitoring progress with real-time analytics.

Azure Utility Tool | *DotNet, C, Azure Resouce Manager, Azure API*

Mar 2023 – May 2023

- Administered a scalable Azure resource monitoring tool using Azure APIs and Terraform, automating cost reporting and reducing operational costs by 20%.

TECHNICAL SKILLS

Languages: Java, Python, C#, C/C++, SQL, JavaScript, Typescript, HTML/CSS

Frameworks: React, Angular, Node.js, Flask, JUnit, WordPress, Material-UI, FastAPI

Developer Tools: Git, Docker, VS Code, Visual Studio, PyCharm, IntelliJ, Eclipse

Libraries: pandas, NumPy, Matplotlib

Databases: MySQL, MongoDB, Firebase

Methodologies: Agile Development, Sprint Planning, CI/CD(Jenkins, Azure Devops)